

Luís Carlos F. Latoski | Curriculum Vitae

Instituto de Física, Universidade Federal do Rio Grande do Sul

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Research interests

Nonequilibrium statistical mechanics and complex systems. Coarsening and interface dynamics, opinion models with memory, stochastic processes on lattices, and percolation in biological networks.

Education

Universidade Federal do Rio Grande do Sul

Porto Alegre, Brazil

PhD candidate in Physics

2022–2026

Opinion models, inertia and coarsening. Supervisor: Prof. Jeferson J. Arenzon (UFRGS). Co-supervisor: Prof. Wellington G. Dantas (UFF).

Universidade Federal do Rio Grande do Sul

Porto Alegre, Brazil

MSc in Physics

2020–2022

O papel de zelotes emergentes no modelo do votante (Role of emergent zealots in the voter model). Supervisor: Prof. Jeferson J. Arenzon (UFRGS). Co-supervisor: Prof. Wellington G. Dantas (UFF).

Universidade Federal do Rio Grande do Sul

Porto Alegre, Brazil

BSc in Physics

2012–2019

Estudo de modelos de rede por meio de algoritmos de execução em paralelo. Advisor: Prof. Heitor C. M. Fernandes (UFRGS).

Publications

2025: L. C. F. Latoski, A. De Martino, D. De Martino. *Cross-feeding percolation phase transitions of intercellular metabolic networks.* **Science Advances** (2025). doi:10.1126/sciadv.adv8216. arXiv:2412.09088.

2024: L. C. F. Latoski, W. G. Dantas, J. J. Arenzon. *Opinion inertia and coarsening in the persistent voter model.* **Phys. Rev. E** **109**, 054115 (2024). doi:10.1103/PhysRevE.109.054115. arXiv:2403.02483.

2022: L. C. F. Latoski, W. G. Dantas, J. J. Arenzon. *Curvature-driven growth and interfacial noise in the voter model with self-induced zealots.* **Phys. Rev. E** **106**, 014121 (2022). doi:10.1103/PhysRevE.106.014121. arXiv:2204.03402.

Selected talks and presentations

2026: *Cross-feeding percolation phase transitions of intercellular metabolic networks.* [simbad] School and Workshop, University of Havana, Cuba, February 2026. Oral.

2025: *Cross-feeding percolation phase transitions of intercellular metabolic networks.* First Simbad Workshop, Turin, Italy, September 2025. Oral.

2025: São Paulo School of Advanced Science on Disordered Systems, ICTP-SAIFR / IFT-UNESP, São Paulo, April–May 2025. Poster.

2024: *Opinion inertia and coarsening in the persistent voter model.* Brazilian Meeting on Statistical Physics, Curitiba, November 2024. Oral.

2024: *Opinion inertia and coarsening in the persistent voter model.* Encontro de Outono da SBF, Florianópolis, May 2024. Oral.

2024: *Percolation of single-cell cross-feeding networks during metabolic overflow.* Encontro Nacional de Física Estatística, 2024.

2023: *Curvature-driven growth and interfacial noise in the voter model with self-induced zealots.* Encontro de Outono da SBF, Ouro Preto, May 2023. Oral.

2021: *Synchronization of epidemic oscillations induced by social network control.* London Mathematical Laboratory Summer School, July 2021. Oral.

2021: *Impact of self-induced disorder on the voter model.* Brazilian Meeting on Statistical Physics, November 2021. Poster.

Schools and workshops attended

2023: Minicourse on Lattice Models and Applications to Biological Problems, ICTP-SAIFR / IFT-UNESP, São Paulo, November–December 2023.

2023: Advanced School on Quantitative Principles in Microbial Physiology, ICTP, Trieste, Italy, October 2023.

2021: Social Phenomena from a Physics Perspective, ICTP-SAIFR, October 2021.

2021: Statistical Physics of Complex Systems, ICTP and EPS, September 2021.

Fellowships

2022–: PhD fellowship, CNPq, full-time.

2020–2022: MSc fellowship, CAPES, full-time.

2019–2020: Undergraduate research fellowship, UFRGS. Advisor: H. C. M. Fernandes.

2015: Outreach fellowship, UFRGS. Primary-school science outreach.

Awards

2019: Distinction, poster session, Computational and Interdisciplinary Physics. XXXI Salão de Iniciação Científica, UFRGS.

2019: Distinction, oral session, Computational and Interdisciplinary Physics. XXXI Salão de Iniciação Científica, UFRGS.

Peer review

2026: Referee, *Humanities and Social Sciences Communications* (Nature Portfolio).

2025: Referee, *EPJ Data Science* (Springer).

Skills

Computing: C, C++, Python (NumPy, SciPy), OpenMP, parallel Monte Carlo, Git, $\text{\LaTeX}$.

Methods: Kinetic Monte Carlo, master equations, finite-size scaling, coarse-graining, percolation analysis.

Languages: Portuguese (native), English (fluent).